**Math 4441** **Probability and Statistics** **Credit: 3.00**

Probability Law: Sets, Probabilistic Models, Conditional Probability, Independence, Total Probability Theorem, Bayes' Theorem, Counting.

Discrete Random variables: Probability Mass Functions (PMF), Cumulative Distribution Functions (CDF), Expectation, Variance; Wellknown distributions (Uniform distribution, Bernoulli distribution, Binomial distribution, Poisson distribution. etc.). Continuous Random variables: Probability Density Functions (PDF), Cumulative Distribution Functions (CDF), Expectation, Variance; Well-known distributions (Uniform distribution, Exponential distribution, Gaussian distribution).

Joint Random Variables: Joint PMFs, PDFs, Conditional Expectation, Covariance, Correlation, Independence of random Variables.

Inferential Statistics and Probability Models, Populations and Samples. Descriptive Statistics: Describing Data Sets, Summarizing Data Sets and Chebyshev's Inequality. The Sample Mean, the Central Limit Theorem, the Sample Variance, Sampling Distributions from a Normal Population. Parameter Estimation: Maximum Likelihood Estimators, Interval Estimates. Hypothesis Testing: Significance Levels, Tests Concerning the Mean of a Normal Population, Hypothesis Tests Concerning the Variance of a Normal Populations. Distribution of the Estimators.

**Recommended Texts:**

1. S. M. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 5th ed. Academic Press, 2014
2. R. D. Yates and D. J. Goodman, *Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers*, 3rd ed. Wiley, 2014

**CSE 4403** **Algorithms** **Credit: 3.00**

Techniques for analysis of algorithms, Methods for the design of efficient algorithms: divide and conquer, greedy method, dynamic programming, back tracking, branch and bound, Basic search and traversal techniques, graph algorithms, Algebraic simplification and transformations, lower bound theory, NP-hard and NP-complete problems.

**Recommended Texts:**

1. T. H. Cormen et al., *Introduction to Algorithms*, 4th ed. The MIT press, 2022
2. E. Horowitz et al., *Fundamentals of Computer Algorithms*, 2nd ed. Universities Press (India) Private Limited, 2008



**CSE 4409** **Database Management Systems II** **Credit: 2.00**

Relational Database Programming: Introduction, its role in S/W development; Relational Database Basic Constructs: Table, Keys, Views, Cardinality; Introduction to SQL, Relational query and subquery, Redundancy and Functional composition in Database; Concept of Joins, Natural joins; Views, its usage and restrictions, Introduction to PL/SQL, PL/SQL Control Structures, Functions and Procedures, Introduction to Cursor, Records, Transaction Management, Oracle Collection, Large Objects, PL/SQL Package, Database Triggers, Dynamic SQL, Introduction to Database Administration, Database Performance Tuning, Brief Introduction to other Relational Databases such as: MySQL, PostGRE, MS SQL Server, Database Security.

**Recommended Texts:**

1. M. McLaughlin, *Oracle Database 11g PL/SQL Programming*. O'Reilly, 2008

**CSE 4411** **Data Communication and Networking** **Credit: 3.00**

Introduction - Internet, Network edge, Network Core, Access Networks, Protocol Layers and Service models, Application Layer - Principles of Network applications, Web, HTTP, FTP, DNS, Socket Programming, Transport Layer - Transport Layer services, Multiplexing and Demultiplexing, Connectionless transport and connection oriented transport, Principles of congestion control, Reliability, Network Layer - Router, Internet Protocols, Routing algorithms, broadcast and multicast routing, Link Layer - Error detection and correction techniques, multiple access protocols, Link Layer addressing, Random Access techniques, Hubs and Switches, Wireless and mobile networks, Multimedia Networking - Streaming of Stored Audio and Video, Protocols of Real time Interactive applications, Physical Layer - Transmission Medium, Encoding and Decoding, Error detection and correction.

**Recommended Texts:**

1. J. F. Kurose and K. W. Ross, *Computer Networking*, 8th ed. Pearson, 2020
2. B. A. Forouzan, *Data Communications and Networking*, 5th ed. McGraw Hill, 2012
3. A. S. Tanenbaum, *Computer Networks*, 6th ed. Pearson, 2020

**SWE 4401** **Software Requirement and Specifications** **Credit: 3.00**

Basics of requirements engineering, types of requirements -functional requirements, non-functional requirements, quality attributes, main requirements engineering activities, documents and processes; Requirements inception and elicitation: product vision and project scope, traditional elicitation approaches (interviews, stakeholders study, workshops, etc), scenario/use case approaches, prototyping, requirements